

Continuous functions – 5

The intermediate value theorem as a topological tool

Homework

1 сентября 2026 года

Theoretical minimum. Connectedness and the intermediate value property; the difference trick; the horizontal chord theorem; Borsuk–Ulam in dimension one and the pancake theorem; Darboux functions; fixed points on a segment; sign changes and counting arguments.

1. Prove that a continuous function on a connected set which takes both positive and negative values vanishes somewhere, and deduce that a polynomial of odd degree has a real root.
2. Prove the horizontal chord theorem for $\alpha = \frac{1}{n}$ and construct, for every α not of this form, a continuous f with $f(0) = f(1)$ and $f(x+\alpha) \neq f(x)$ everywhere.
3. Prove that a continuous closed curve in the plane which is not a single point admits an inscribed rhombus, and state the inscribed square problem together with what is known about it.
4. Let $f, g: [0, 1] \rightarrow [0, 1]$ be continuous with $f \circ g = g \circ f$. Prove that if f is monotone, then f and g have a common fixed point; give the standard counterexample showing that the statement fails for general continuous f and g .
5. Prove that a continuous map of a circle to itself of degree zero has a fixed point, and give an example of a fixed-point-free map of the circle.
6. Two runners start at the same point of a circular track and run with continuous speeds, finishing at the same point after one hour. Prove that at some moment they are diametrically opposite, provided their total distances differ by an odd multiple of half of the track.

7. Prove that every convex bounded region in the plane admits a line which bisects both its area and its perimeter.
8. Prove that for a continuous $f: [0, 1] \rightarrow \mathbb{R}$ with $f(0) = f(1)$ the set of chord lengths $\{\alpha : \exists x, f(x + \alpha) = f(x)\}$ is closed and contains $\frac{1}{n}$ for every n .
9. Give an example of a Darboux function which is discontinuous everywhere, and prove that a Darboux function which is injective is continuous.
10. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous and suppose that f is unbounded above and below on every circle centred at the origin. Prove that the zero set of f meets every such circle, and deduce that it is unbounded.